

What is in the Waterbug Testing Kit?



Dissecting
Microscope



Bulb Pipettes



6-dish plate



Net



Gumboots



Spoons and
Ladle



White Tubs



Bucket

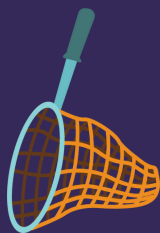
How to Collect your own waterbug samples from a waterway

1 Entering a Waterway (always with supervision from an adult)

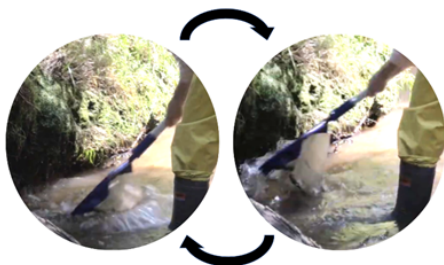
- DO NOT enter if the water is too deep.
- DO NOT enter if water is moving fast.
- Find the safest spot to enter the water.
- Use your net to check the depth and look for any hidden sinkholes.
- Stay in shallow areas



2 Collect Samples



Use the net to collect water in the water trays with one of these methods



To collect bugs near the edges or plants, place your net in the water and scoop it up



To collect bugs near rocks or in open areas, hold your net in front of you in the water and slowly walk backwards

How to use the Waterbug Testing Kit to conduct an investigation

1 Separate the Waterbugs

- Use spoons to collect waterbugs into a six-dish tray. Separate individual bugs into their own section.



2 Identify the Waterbugs

1. Look at the Waterbugs under the dissecting microscope



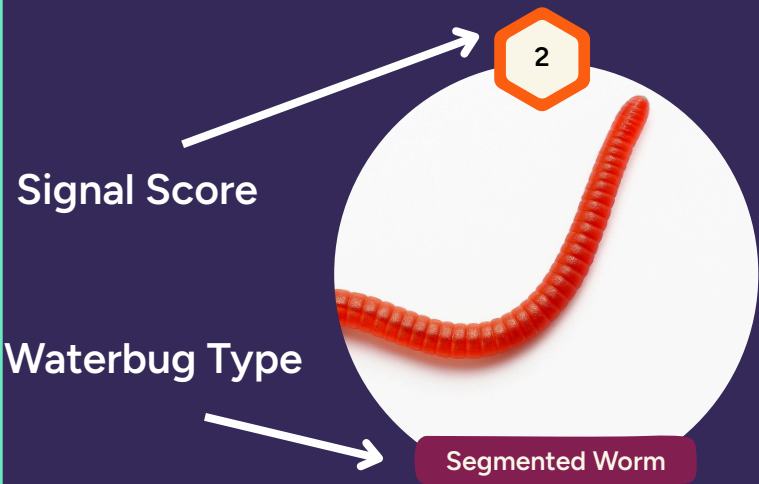
2. Use GTAC's Waterbug ID Resource to identify each waterbug



<https://tinyurl.com/mnryx62u>

3 Record the Signal Score of each waterbug

For each waterbug you find, record what type of bug it is, how many times you see each bug (tally) and its signal score (number from 1 to 10). Use the table provided by GTAC.



Keep count how many times you see the same type of bug

Type of Waterbug	Tally	Signal Score
Segmented Worm	III	2

4 Calculate totals

Calculate the total types of waterbugs you saw (blue box) and the total signal scores (green box)

Type of Waterbug	Tally	Signal Score	Tally x Signal Score
Dragonfly Nymph	I = 1	3	3 x 1 = 3
Water Strider	II = 2	4	4 x 2 = 8
	Total Tally: 3		Total Signal Score: 3 + 8 = 11